

14_November_2025_provisional • Unit 2

1H + 2H source-order past paper

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REVIEWED PROVISIONAL | L-to-M

Converted from Linear Higher papers; not an official Pearson paper or mark scheme.

Contents

- 1H Q01 • Statistics Toolkit • 4 marks
- 1H Q03 • Bearings, Scale Drawing & Constructions • 2 marks
- 1H Q04 • Statistics Toolkit • 3 marks
- 1H Q05 • Algebraic Proof • 3 marks
- 1H Q06 • Ratio Problem Solving • 4 marks
- 1H Q09 • Percentages • 3 marks
- 1H Q12 • Angles in Polygons & Parallel Lines • 4 marks
- 1H Q16 • Statistics Toolkit • 2 marks
- 1H Q17 • Algebraic Proof • 3 marks
- 1H Q18 • Rearranging Formulas • 4 marks
- 1H Q24 • Transformations of Graphs • 3 marks
- 1H Q25 • Solving Inequalities • 3 marks
- 1H Q27 • Vectors • 6 marks
- 1H Q28 • Coordinate Geometry • 6 marks
- 2H Q02 • Simultaneous Equations • 3 marks
- 2H Q04 • Congruence, Similarity & Geometrical Proof • 3 marks
- 2H Q05 • Compound Interest & Depreciation • 6 marks
- 2H Q08 • Powers, Roots & Standard Form • 3 marks
- 2H Q10 • Graphing Inequalities • 4 marks
- 2H Q12 • Graphs of Functions • 5 marks
- 2H Q14 • Cumulative Frequency Diagrams • 5 marks
- 2H Q15 • Circle Theorems • 4 marks
- 2H Q16 • Algebraic Proof • 2 marks
- 2H Q17 • Direct & Inverse Proportion • 3 marks
- 2H Q19 • Functions • 3 marks
- 2H Q20 • Forming & Solving Equations • 3 marks
- 2H Q22 • Differentiation • 6 marks
- 2H Q23 • Area & Volume of Similar Shapes • 3 marks
- 2H Q25 • Sequences • 5 marks
- 2H Q26 • Standard & Compound Units • 6 marks

- 1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

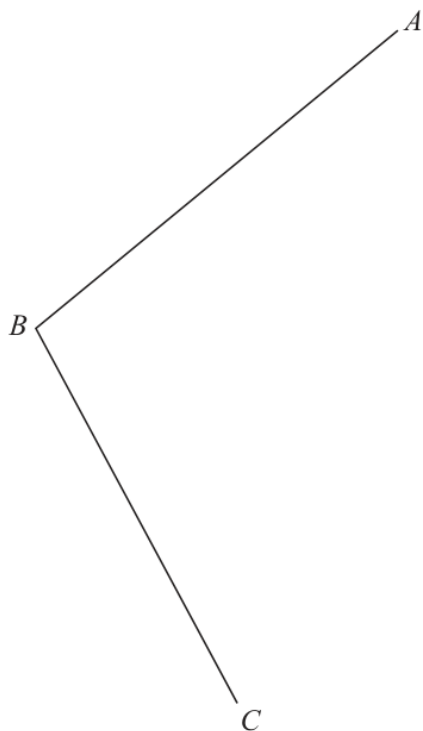
Number of kilometres (K)	Frequency
$0 \leq K < 5$	8
$5 \leq K < 10$	7
$10 \leq K < 15$	3
$15 \leq K < 20$	10
$20 \leq K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

..... kilometres

(Total for Question 1 is 4 marks)

- 3 Using ruler and compasses only, construct the bisector of angle ABC
Show all your construction lines.



(Total for Question 3 is 2 marks)

4 e, f and g represent integers such that

$e \quad f \quad 7 \quad g \quad 12 \quad 17$

is a list of integers written in order of size.

The integers have

- a range of 15
- a median of 8.5
- a mean of 9

Work out the value of e , the value of f and the value of g

$e = \dots\dots\dots$

$f = \dots\dots\dots$

$g = \dots\dots\dots$

(Total for Question 4 is 3 marks)

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

- 6 Eli and Peta share \$275 in the ratio 2 : 3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$.....

(Total for Question 6 is 4 marks)

- 9 Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.
Work out the number of points that Hari scored playing the game on Monday.

(Total for Question 9 is 3 marks)

12

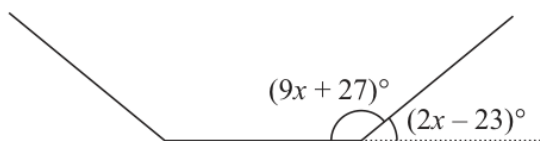


Diagram **NOT**
accurately drawn

The diagram shows part of a regular n -sided polygon with

an interior angle of $(9x + 27)^\circ$

an exterior angle of $(2x - 23)^\circ$

Work out the value of n

$n = \dots\dots\dots$

(Total for Question 12 is 4 marks)

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

(Total for Question 16 is 2 marks)

- 17 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

18 Make y the subject of $p = \sqrt{\frac{y+w}{3y-t}}$

(Total for Question 18 is 4 marks)

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)
(1)

(ii) $y = f(x) + 7$

(..... ,)
(1)

(iii) $y = f(2x)$

(..... ,)
(1)

(Total for Question 24 is 3 marks)

- 25** Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

27

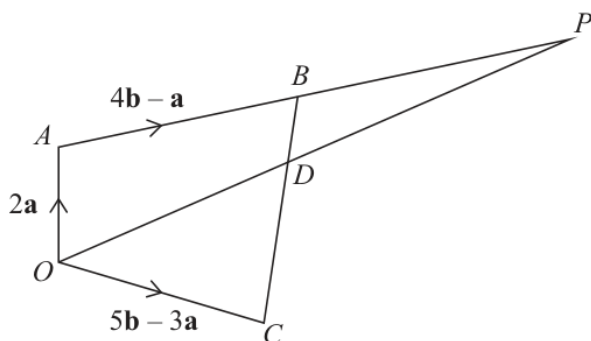


Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$
Simplify your answer.

(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n



$$n = \dots\dots\dots (4)$$

(Total for Question 27 is 6 marks)

- 28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are $(0.2, 2.6)$

Work out the coordinates of B
Show your working clearly.



(..... ,)

(Total for Question 28 is 6 marks)

2 Solve the simultaneous equations

$$\begin{aligned}5x + y &= 11 \\ 3x - y &= 9\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 2 is 3 marks)

- 4 Here are three similar quadrilaterals.

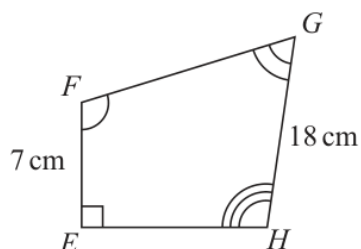
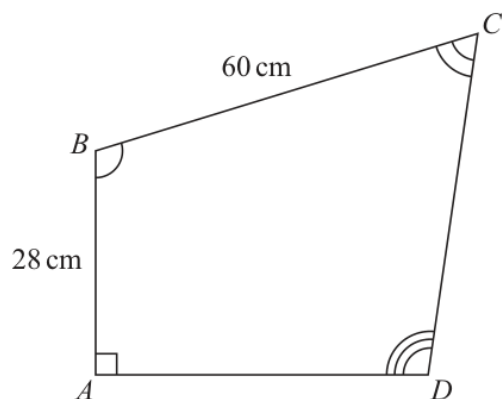
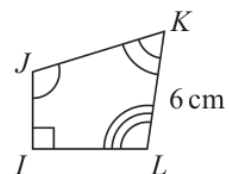


Diagram **NOT** accurately drawn



Work out the length of JK

..... cm

(Total for Question 4 is 3 marks)

5 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.

On Saturday, Otis sells 87 ice creams.

- (a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %
(3)

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

- (b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs
(3)

(Total for Question 5 is 6 marks)

- 8 (a) Write 5.76×10^4 as an ordinary number.

.....
(1)

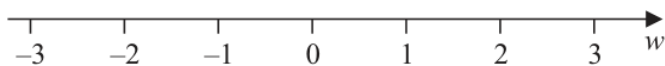
(b) Work out $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

.....
(2)

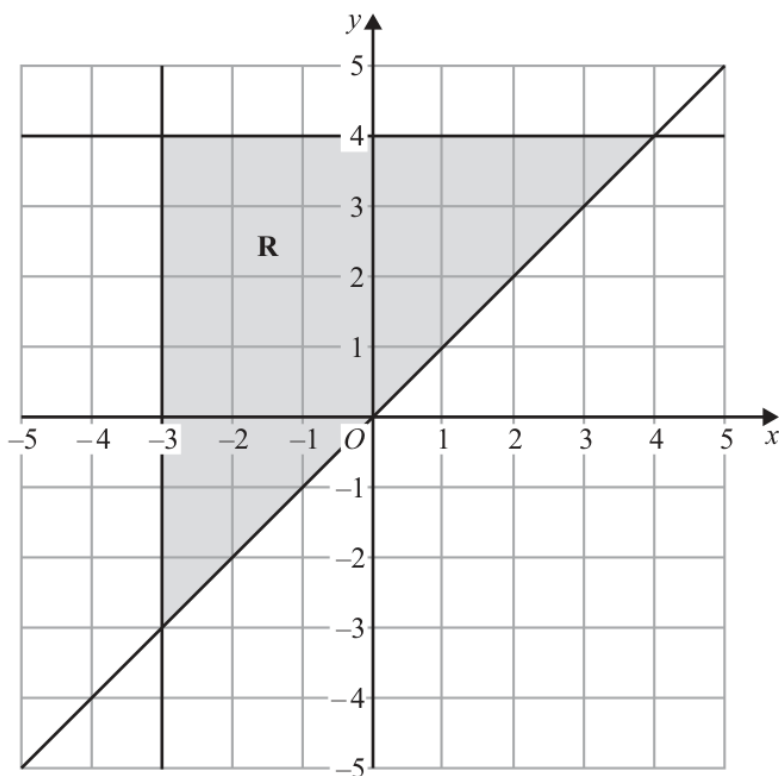
(Total for Question 8 is 3 marks)

- 10 (a) On the number line, represent the inequality $w < 1$



(1)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....

.....

.....

(3)

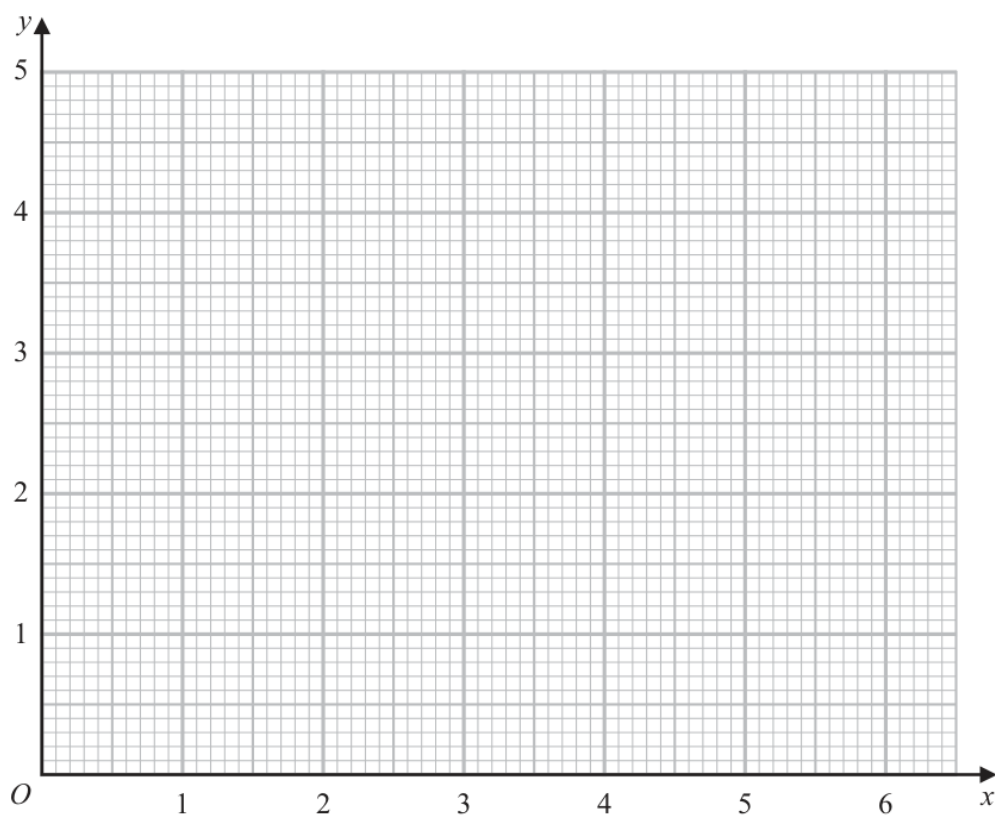
(Total for Question 10 is 4 marks)

12 (a) Complete the table of values for $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$

x	0.5	1	2	3	4	5	6
y	4.25			2.17	2.5	2.9	3.33

(1)

(b) Draw the graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ for $0.5 \leq x \leq 6$



(2)



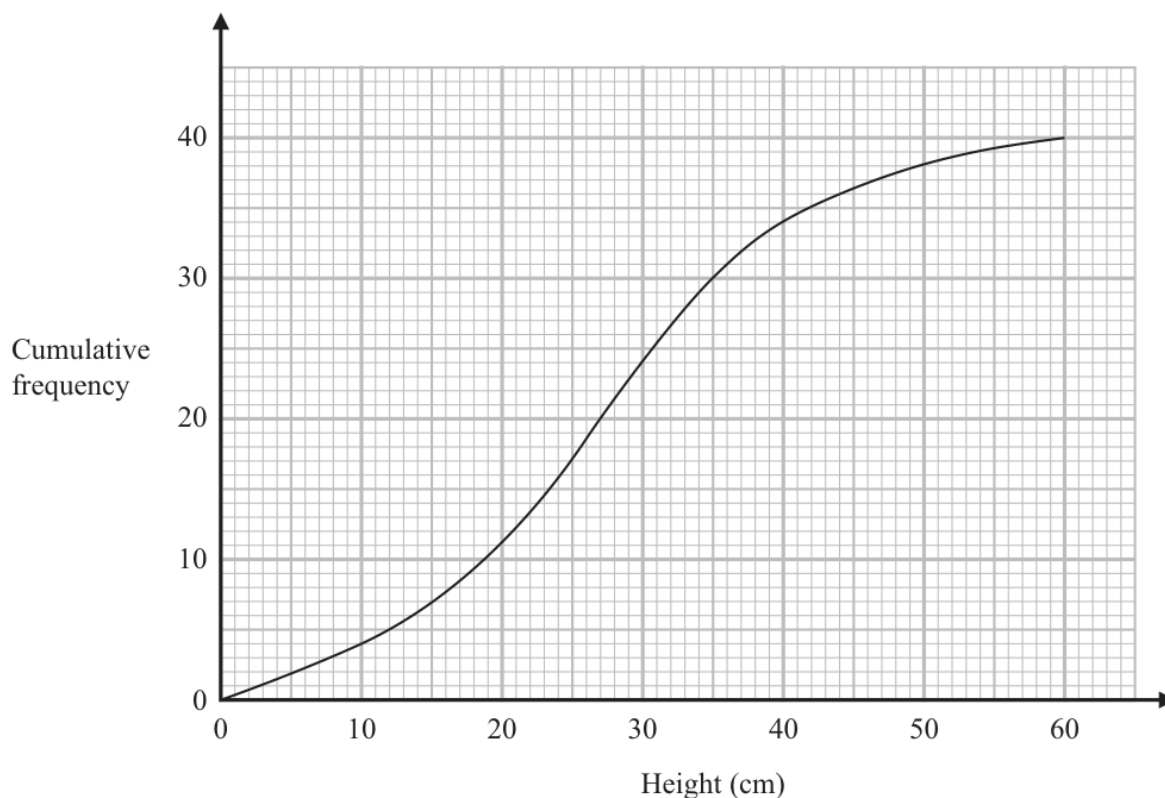
- (c) By drawing a suitable line on the grid, find estimates for the solutions of the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

.....
(2)

(Total for Question 12 is 5 marks)

- 14 The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.



- (a) Use the graph to find an estimate for the median height.

..... cm
(1)

- (b) Use the graph to find an estimate for the interquartile range of the heights.

..... cm
(2)



Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

(c) Work out the total amount of money Greta receives for the premium plants.

..... euros
(2)

(Total for Question 14 is 5 marks)

15 A, B, C and D are points on a circle with centre O

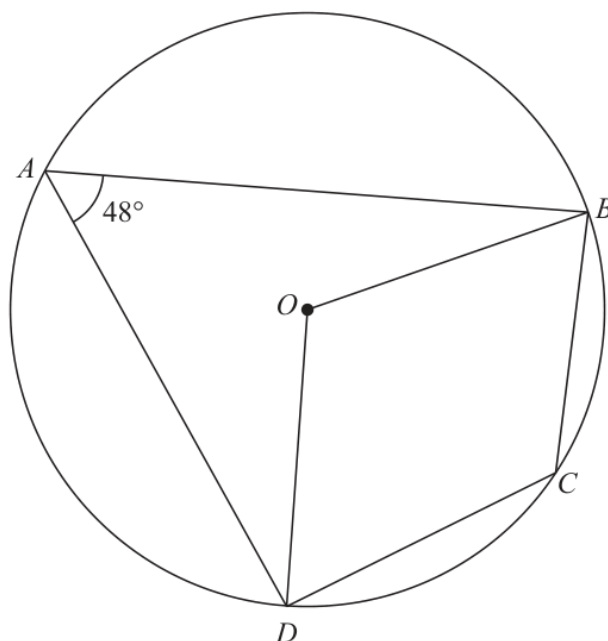


Diagram **NOT**
accurately drawn

Angle $DAB = 48^\circ$

(a) (i) Work out the size of the obtuse angle DOB

o

(1)

(ii) Give a reason for your answer.

(1)

(b) (i) Work out the size of angle BCD

o

(1)

(ii) Give a reason for your answer.

(1)

(Total for Question 15 is 4 marks)

16 Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

(Total for Question 16 is 2 marks)

17 T is inversely proportional to $\sqrt{m}$

$$T = 15 \text{ when } m = 36$$

Find a formula for T in terms of m

(Total for Question 17 is 3 marks)

19 The functions f and g are such that

$$f(x) = 3x - 2$$

$$g(x) = \frac{x}{2x - 1}$$

(a) Find $g(3)$

.....
(1)

(b) Find $gf(x)$

Give your answer in its simplest form.

$gf(x) =$
(2)

(Total for Question 19 is 3 marks)

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

$a =$

$c =$

(Total for Question 20 is 3 marks)

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**

..... cm³

(Total for Question 23 is 3 marks)

- 25** The first term of an arithmetic series is 10
The 20th term of the series is 86
- The sum of the first N terms of the series is 5194
- Work out the value of N
Show your working clearly.

$N = \dots\dots\dots$

(Total for Question 25 is 5 marks)

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

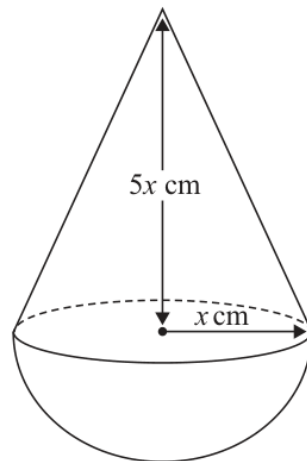


Diagram **NOT**
accurately drawn

The cone is made from copper.
The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams
The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.
Show your working clearly.



..... g/cm³

(Total for Question 26 is 6 marks)